TASK 4: POLYMORPHISM

Python OOP Practical Report — 15 Marks

1. Objective

To demonstrate polymorphism using two different classes that have a method with the same name but perform different actions.

2. OOP Concept Used

Polymorphism, same method name, and different behavior for different objects.

3. Source Code

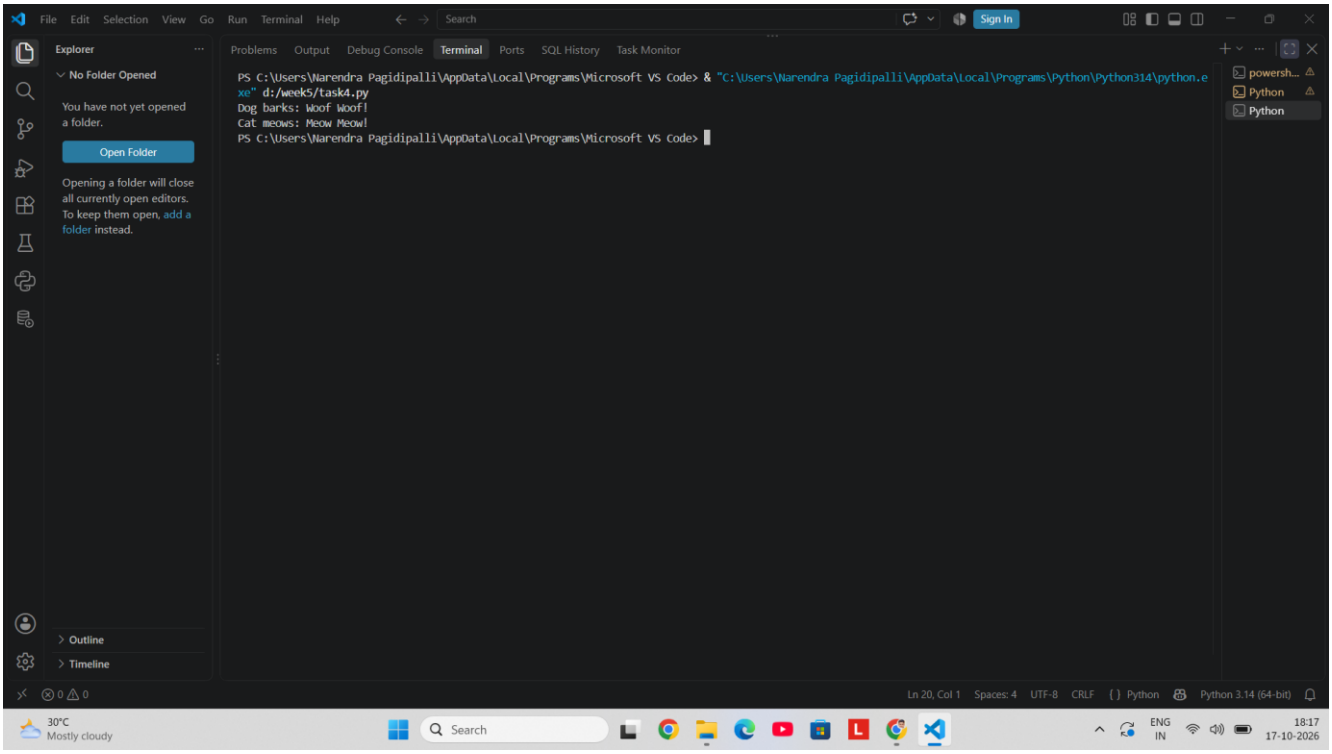
```
class Dog:
    def sound(self):
        print("Dog barks: Woof Woof!")


class Cat:
    def sound(self):
        print("Cat meows: Meow Meow!")


d = Dog()
c = Cat()

d.sound()
c.sound()
```

4. Output



5. Explanation

- Dog and Cat are two different classes.
- Both classes define a method named sound().
- The Dog object's sound() method prints a barking sound.
- The Cat object's sound() method prints a meowing sound.
- The same method call name produces different outputs depending on the object.

6. Learning Outcome

I learned that different classes can define the same method name and implement different behaviors.

7. Conclusion

This program demonstrates polymorphism through the sound() method in the Dog and Cat classes.